Latest updates: https://dl.acm.org/doi/10.1145/3012430.3012556

RESEARCH-ARTICLE

Latest technology trends in health sciences education

JUAN ANTONIO JUANES, University of Salamanca, Salamanca, Salamanca, Spain

PABLO RUISOTO, University of Salamanca, Salamanca, Salamanca, Spain

Open Access Support provided by:

University of Salamanca

PDF Download
3012430.3012556.pdf
30 March 2026
Total Citations: 2
Total Downloads: 180

Published: 02 November 2016

Citation in BibTeX format

TEEM'16: 4th International Conference on Technological Ecosystems for Enhancing Multiculturality
November 2 - 4, 2016
Salamanca, Spain

TEEM '16: Proceedings of the Fourth International Conference on Technological Ecosystems for Enhancing Multiculturality (November 2016)
https://doi.org/10.1145/3012430.3012556
ISBN: 9781450347471

Latest Technology Trends in Health Sciences Education

Juan Antonio Juanes
Visual Med System Research Group. Educational Sciences Research Institute
University of Salamanca,
Spain
jajm@usal.es

Pablo Ruisoto
Visual Med System Research Group
University of Salamanca,
Spain
ruisoto@usal.es

ABSTRACT

The purpose of this track is to present a general overview of contributions made by computer and medical technology applications and related fields in health sciences training; providing an excellent opportunity for the promotion and experience exchange of innovative teaching experiences, especially those that use the latest technology. With this in mind, the Track includes papers on the latest tendencies of innovative teaching, in both methodology and technology (clinical simulation, medical imaging treatment and manipulation software, biomedical databases for training in computer aided diagnosis, software solutions and medical image processing platforms, artificial neural networks, deep learning methods based on learning data representations, medical training applications for mobile devices such as smartphones and tablets, as well as advanced medical visualization systems with virtual reality techniques using stereoscopic vision headsets, among others), all of them used in health sciences education, through the experience of different research groups.

CCS Concepts

• Human-centered computing→Human computer interaction (HCI)

• Life and medical sciences→Health care information systems

• Life and medical sciences→Health informatics

Keywords

Immersive system; stereotypic glasses; human anatomy; virtual reality virtual.

1. LATEST TECHNOLOGY TRENDS IN HEALTH SCIENCES EDUCATION

Technology applied to health sciences education has undergone major changes over time, always parallel to the technological evolution of society. Technological advances have thus changed teaching and learning methods among university students in experimental and health sciences specialties.

The relationship between the current information society context, Medicine and healthcare systems is especially important given that in these fields information and knowledge are particularly important. The different devices that the technology industry creates every year, in rapid growth, are incorporated and used in our society on a daily basis. Therefore, the impact that information and communication technologies have on health sciences education make it possible for products to be developed and incorporated regularly in teaching. The application of computer applications in experimental and health sciences education broadens the possibilities for action and communication between teachers and students. It also allows access to new means of information exploration and representation and to knowledge through different types of resources, such as computer simulations, stereoscopic vision systems with augmented reality glasses, computer platforms for the management and use of resources and documents, etc.

Knowledge transfer, regardless of space and time, using technology and simulation systems with virtual resources, is nowadays one of the greatest technological, scientific and cultural revolutions of our society, and the health sciences have managed to take advantage of it.

The ability to understand body structures is closely related to the ability to visualize them. Traditionally, the different imaging techniques have shared a common limitation, the two-dimensionality of their representations. The visualization and interpretation of human body images obtained from these techniques is an added difficulty due to their complexity and morphological individual differences (biological variability). Learning human anatomy, one of the most important disciplines in health sciences degrees, requires an understanding of the morphological characteristics and spatial relations between the different body structures. The volumetric visualization of these structures therefore optimizes the presentation of complex information found within the morphological images using diagnostic imaging techniques. Not only does the visualization capacity improve, but the degree of interaction and realism as well, overcoming the inherent restrictions of traditional 2D representations.

Considering this important aspect of spatial vision, many technology procedures that allow body image reconstruction in three dimensions from sections generated by diagnostic imaging techniques, have emerged in the market. The process of 3D reconstruction or volumetric generation consists in acquiring one or more series of tomographic images, the posterior segmentation or identification and labelling of body structures that are going to be reconstructed, extracting contours / surfaces in adjacent cross sections and rendering the 3D model.

The use of visual resources in 3D format is an attractive and innovative method in teaching and student learning, being reusable and useful tools to encourage and motivate student learning in any health science discipline.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.
TEEM'16, November 02-04, 2016, Salamanca, Spain
© 2016 ACM. ISBN 978-1-4503-4747-1/16/11...$15.00
DOI: http://dx.doi.org/10.1145/3012430.3012556

3D models have not only maximized the resources of the traditional pedagogical system, but renovated it, offering a complete perspective of body structure anatomical features. These resources use visual aids as a backbone, beyond the simple illustrations of textbooks. The generation of three-dimensional images with the help of 3D reconstruction commercial software are part of new visual teaching strategies in order to activate and review contents to better understand body structures.

The visualization of 3D anatomical models creates a more precise and real perspective of the different body structures than that of conventional medical images from 2D diagnostic imaging techniques. Volumetric visualization provides complete information to define the macroscopic morphological features and spatial relations among body structures such as bones or muscles.

On the other hand, 3D models seen in simulated contexts increase the possibilities of making a global assessment of the morphology, and reduce complications in surgical interventions. This advantage is especially important when planning virtual surgical approaches where a precise delimitation of the body structure is needed for the intervention. In recent years, combining volumetric visualization with virtual reality systems has been possible, allowing the surgeon to navigate through the body structures reconstructed in three dimensions.

Stereoscopic vision systems are useful in Medicine since they eliminate some mistakes and help improve the perception and evaluation in clinical practice. This technology plays an important role in minimally invasive surgery to carry out some interventions. The use of virtual reality glasses creates an immersive environment, transforming the images each eye perceives in real time into one stereoscopic image, with good depth perception and excellent technical quality. Stereo vision gathers three-dimensional visual information and creates the illusion of depth in an image, improving medical training through these immersive systems. It is a very useful teaching resource in learning and acquiring skills and competencies necessary for professional practice.

The uses of Virtual Reality (VR) and Augmented Reality (AR) in health sciences have been very diverse. Currently, they are used mainly in surgical simulation through 3D vision systems with stereo vision glasses or headsets. Several technical aspects still need to be worked on so the perception our brain has is complete (like eliminating some physical discomfort when using virtual reality glasses), however these type of spatial vision resources can certainly contribute to the improvement of medical training processes.

VR techniques are now increasingly used in medical training through different applications. VR is a computer system used to create an artificial world, in which the user has the feeling of being inside it and the ability to navigate and manipulate the objects within it. By using a spatial visualization system, the user can have a direct view of the surrounding physical environment where the images generated by the computer are shown. The interactive and immersive visual aids for learning are an incentive for the students and improve their long-term memory.

The use of these new technologies and tools in teaching represents a challenge for teachers who have to adapt to new environments, but at the same time helps achieve the objectives established in the different health sciences degrees.

Technological environments encourage the development of useful tools in university education, including additional and complementary resources which generate different training scenarios, with no time and space restrictions that face to face teaching enforces.

Nowadays, the progressive miniaturization of components has allowed computers to be smaller in size, which facilitates their transportation. In recent years, we have witnessed a revolution in universities with the emergence of mobile devices and applications, currently occupying a prominent position in university education, used by students as complementary tools in their learning process. This tendency has encouraged the appearance of more sophisticated mobile devices that allow the use of interactive apps when teaching, giving students new ways to communicate and get additional information to complement their studies. There is an increasing number of students that go to different information sources using their mobile devices and tablets. The growth of these resources for portable devices is playing an important role to improve the quality and efficiency of education processes.

The contents being developed for these mobile devices are usually conceived as a complement for both face to face and online teaching. Many mobile phone companies have professional teams designing training programs in different knowledge areas. The final aim with the use of these technologies is to teach students the basic skills to learn independently.

Mobile devices can be effective tools for students to manage their learning process as a complement to traditional resources. Nowadays there are a number of applications for mobile phones, whose implementation in university teaching is an excellent means to acquire knowledge and evaluate independently, actively and responsibly, to acquire skills in health sciences. The use of these technologies makes students go from being mere passive recipients of content, to creative participants in constant communication and interaction with their learning process.

The use of technology systems in clinical practice facilitate the training in different medical techniques, acquiring basic skills for clinical explorations like cardiopulmonary auscultation, ultrasound examination, or even more complex ones to perform an endoscopy, laparoscopy, etc. Recently, technological applications are being developed to learn practical skills by using manual control devices. They are useful tools in the process of medical training to develop and acquire clinical skills. A small device located in front of the monitor can be connected to the computer through a USB cable, able to capture our hand and finger movements with high precision. This possibility allows the creation of tools using the latest technology to practice clinical procedures manually and develop skills through constant training. These interaction devices have monochromatic cameras and infrared LEDs, interacting simply and intuitively with the computer by using our hands. A carefully designed virtual experience can certainly give the user (student) a sense of control over the environment, even though it is fictional, and facilitate the learning and training processes in Medicine. These developments in simulation technology provide a unique access to learning resources in daily medical training and clinical practice. Currently, an increasing number of computers on the market integrate motion sensor technologies. In the not too distant future, there will certainly be changes and new features added to improve this technology.

One of the biggest challenges of any knowledge area, in an information and knowledge technological society, is the search for solutions to problems present in daily clinical practice. Therefore, the use of computer simulation technologies is a means to present information in an attractive way and close to reality that the university student and teacher will have to deal with.

It is clear that technological advances define the basis of modern education and give us the possibility of using it in health sciences education. These resources offer a more versatile approach to knowledge representation, without losing the classical descriptive education. Clinical simulation rooms are being increasingly used with virtual patients in the shape of high quality mannequins, on which clinical procedures and techniques can be carried out like real patients. This technology provides excellent support materials for medical training, benefiting the learning process and practical skills acquisition by training with these simulators, close to the reality that students will certainly face in their future.

One of the latest technology trends is the use of anatomical materials reconstructed with 3D printers. As a result of their boom, along with a reduction of printer production costs, and the possibility of using their open source software, they have expanded and aroused great interest in many medical specialties, especially in education to create anatomical parts useful for teaching human anatomy. This subject requires the use of human study for practice. However, this material is frequently deteriorated and does not display many details of body structures. The most characteristic example are bones. They are often broken and lacking relevant details. The three-dimensional impression of this material facilitates and complements teaching in the classroom. This type of technology, in healthcare, is allowing the development of anatomical parts to scale of a specific body part.

Currently, 3D printing technology is at a point of maturity that could lead to a significant expansion in the productive system, due to lower equipment prices and the evolution of knowledge about this technology and new materials. The design of human body anatomical structures in three dimensions can help students have a better understanding of their morphological details, in addition to being a long-lasting teaching material.

Recently, many research projects based on the creation of artificial human body parts in order to replace organs damaged by accidents or serious illness are emerging. In the not too distant future, these 3D printers will possibly far exceed the problems that they may pose nowadays; surely, this revolutionary technology will continue to evolve in the field of Medicine.

Technological advances in health sciences are evidently benefiting both medical training and clinical practice.

The use of technological environments in medical training, used always as additional resources in teaching, facilitates and optimizes learning and acquiring clinical skills in students and residents of different health sciences specialties.

In recent years, the development of computer applications has grown significantly, not only in number, but in the different fields that have benefited from them. In health sciences education in general, and Medicine in particular, the progressive incorporation of technological advances has transformed the learning and teaching process, leading to "educational technology" [1-18].

The use of technologies in health sciences education has radically changed the traditional teaching conditions due to new teaching and learning approaches. These changes in health science teaching have represented a challenge for teachers and institutions when designing, developing, implementing and evaluating these applications, as well as an effort to adapt by students. Today more than ever, health science teachers need to be involved in learning and mastering emergent technologies, considering the rapid growth of technological systems and devices that are present in our society. It is thus fitting to provide a space to put forth and reflect on these practices at the International Conference on Technology Ecosystems for Enhancing Multiculturality, and in the conference, this track on Latest Technology Trends in Health Science Education. It is a good representation of their impact and current diffusion among the different specialties of health sciences education.

2. REFERENCES

[1] Abreu García, M., Regalado Miranda, E., and Roque Acosta, M.C., 2001. Nuevas tecnologías de la información y la comunicación en la enseñanza de las Ciencias Médicas. *Educación Médica Superior 15*, 3, 279-283.

[2] Álvarez Sánchez, D. and Edwards Schacthet, M., 2006. El teléfono móvil: una herramienta eficaz para el aprendizaje activo. In *Current Developments in Technology-Assisted Education*, A. Méndez-Vilas, A.S. Martín, J.a.M. González and J.M. González Eds. Formatex, Badajoz, Spain, 23-30.

[3] Briz Ponce, L. and García-Peñalvo, F.J., 2015. An empirical assessment of a technology acceptance model for apps in medical education. *Journal of Medical Systems 39*, 11, Paper 176. DOI= http://dx.doi.org/10.1007/s10916-015-0352-x.

[4] Briz-Ponce, L. and Juanes Méndez, J.A., 2015. Mobile Devices and Apps, Characteristics and Current Potential on Learning. *Journal of Information Technology Research 8*, 4, 26-37. DOI= http://dx.doi.org/10.4018/JITR.2015100102.

[5] Briz Ponce, L., Juanes Méndez, J.A., and García-Peñalvo, F.J., 2015. Dispositivos móviles y apps: Características y uso actual en educación médica. *Novática. Revista de la Asociación de Técnicos en Informática 231*, 86-91.

[6] Briz-Ponce, L., Juanes-Méndez, J.A., and García-Peñalvo, F.J., 2015. Synopsis of discussion session on defining a new quality protocol for medical apps. In *Proceedings of the Third International Conference on Technological Ecosystems for Enhancing Multiculturality (TEEM'15) (Porto, Portugal, October 7-9, 2015)*, G.R. Alves and M.C. Felgueiras Eds. ACM, New York, NY, USA, 7-12. DOI= http://dx.doi.org/10.1145/2808580.2808582.

[7] Briz-Ponce, L., Juanes-Méndez, J.A., García-Peñalvo, F.J., and Pereira, A., 2016. Effects of Mobile Learning in Medical Education: A Counterfactual Evaluation. *Journal of Medical Systems 40*, 6, Paper 136. DOI= http://dx.doi.org/10.1007/s10916-016-0487-4.

[8] Briz-Ponce, L., Juanes-Méndez, J.A., and García-Peñalvo, F.J., 2016. Handbook of Research on Mobile Devices and Applications in Higher Education Settings IGI Global, Hershey, PA. DOI= http://dx.doi.org/10.4018/978-1-5225-0256-2.

[9] Briz-Ponce, L., Pereira, A., Carvalho, L., Juanes-Méndez, J.A., and García-Peñalvo, F.J., 2017. Learning with mobile technologies – Students' behavior. *Computers in Human*

Behavior In Press. DOI= http://dx.doi.org/http://dx.doi.org/10.1016/j.chb.2016.05.027.

[10] Farrel Vázquez, M.T., 2002. El desafío de las nuevas tecnologías de la información y las comunicaciones para los docentes de la Educación Médica. *Educación Médica Superior 16*, 1, 37-46.

[11] Fernández Gutiérrez, C.F., 2002. Las nuevas tecnologías de la información y las comunicaciones en salud. *Educación Médica Superior 16*, 2, 128-139.

[12] Häfner, P., Häfner, V., and Ovtcharova, J., 2013. Teaching Methodology for Virtual Reality Practical Course in Engineering Education. *Procedia Computer Science 25*, 251–260. DOI= http://dx.doi.org/10.1016/j.procs.2013.11.031.

[13] Juanes Méndez, J.A., 2011. Nuevos entornos tecnológicos para la simulación en educación médica. *Educación Médica 14*, 18-20.

[14] Juanes Méndez, J.A., 2013. Using Smartphones as tools for teaching innovation and training support. In *Actas de las Segundas Jornadas de Innovación Didáctica en la Universidad de Salamanca*, D. Pereira Gómez, J.A. Rodríguez Sánchez and J.L. Heras Santos Eds. Universidad de Salamanca, Salamanca, 111-116.

[15] Juanes Méndez, J.A., 2016. Current status of new technologies in the teaching of experimental sciences and medicine in particular. *Educación Médica 17*, 1, 1-2. DOI= http://dx.doi.org/10.1016/j.edumed.2016.03.001.

[16] Juanes, J.A., Gómez, J.J., Peguero, P.D., and Ruisoto, P., 2016. Digital Environment for Movement Control in Surgical Skill Training. *Journal of Medical Systems 40*, 6, 133. DOI= http://dx.doi.org/10.1007/s10916-016-0495-4.

[17] Juanes, J.A. and Ruisoto, P., 2015. Computer Applications in Health Science Education. *Journal of Medical Systems 39*, 9, 97. DOI= http://dx.doi.org/10.1007/s10916-015-0283-6

[18] Liu, T.C., Wang, H.Y., Liang, J.K., Chan, T.W., Ko, H.W., and Yang, J.C., 2003. Wireless and mobile technologies to enhance teaching and learning. *Journal of Computers Assisted Learning 19*, 3, 371–382. DOI= http://dx.doi.org/10.1046/j.0266-4909.2003.00038.x.